

Bulk Marketplace + MandatePay: System Spec and Team Plan

Event: Wema Hackaholics 7.0 · **Build:** Wed 7 to Thu 8 Oct 2026 · **Pitch:** Fri 9 Oct 2026 **Version:** 1.0 · **Status:** DRAFT until the contracts PR merges (section 5), then contracts are LOCKED. **Platform name:** TBD. Designer proposes three options by Thu 09:00 and the team votes. Until then we say "the Marketplace". **MandatePay** is fixed.

How to use this doc. Section 0 is the 60-second version. Everyone reads sections 1 to 5 once, then jumps to their own part of section 6. The machine-readable files in `/contracts` are the source of truth. If this doc and a contract file disagree, **the file wins** and you open a PR to fix the doc.

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0. Start here

What we are building. Small retailers in one locality pool their orders and buy in bulk directly from producers, cutting out middlemen and unit cost. Each retailer keeps an inventory record. An AI restock agent forecasts demand, recommends purchases and places them. **MandatePay** sits between that agent and the retailer's money: the agent can pay only inside rules the owner signed, risky payments wait for the owner's signed approval, and every action is logged.

Design principle: assume the agent will be fooled. The worst case is the cap, not the balance.

Person	Owns	First deliverable
Backend	<code>services/gateway</code> (MandatePay core)	Auth, signing and mandate endpoints passing the test vector (Wed night)
Software developer	Repo, deploy, <code>services/market</code> , retailer app	Repo scaffold, <code>make dev</code> , mock servers, deploy skeleton (Wed night)
ML engineer	<code>services/agent</code> : forecast, restock agent, attack runner	Sales generator, then forecast with backtest (Wed night)
Frontend	<code>apps/web</code> MandatePay console and shared UI kit	App scaffold, typed client, in-browser signing (Wed night)
Designer	Design tokens, hero screens, pitch deck , demo script	Real price quotes tonight, tokens and 4 hero screens by Thu 09:00

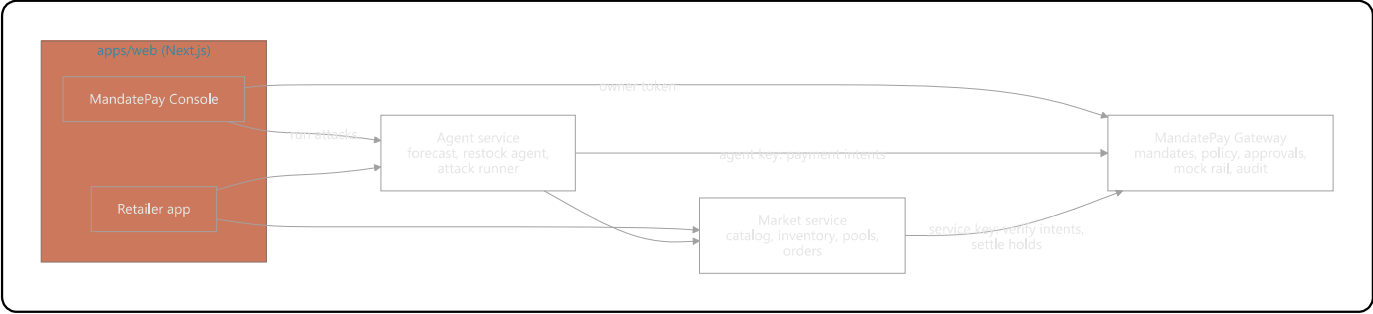
Three rules

1. **Contracts first.** Build against `/contracts` and the mock servers (`make mocks`), never against a teammate's unfinished code.

- 2. **main** always runs. Small branches, PR, merge within a couple of hours.
- 3. **After the lock (target Wed 22:00) contract changes are additive only.** New optional fields are fine. Renames and removals are not. Anything breaking needs all five thumbs-up in the group chat.

1. System overview

1.1 Components



Dependencies run one way: Agent and Market call the Gateway; the Gateway calls nobody. The agent holds no bank credentials, only an agent key that lets it *propose* payments. Only the Gateway touches the (mock) rail.

Component	Path	Port	Owner	Stack (proposal)
Gateway	services/gateway	8001	Backend	Python 3.11, FastAPI, SQLAlchemy 2 + Alembic, Postgres 16
Market	services/market	8002	Software dev	same
Agent	services/agent	8003	ML	Python, FastAPI, pandas/statsmodels, any tool-calling LLM behind an interface
Web	apps/web	3000	Frontend (console), Software dev (retail)	Next.js App Router, TypeScript, Tailwind, shadcn/ui
Mock servers	make mocks	4010 / 4011	Software dev	Prism, from the OpenAPI files

The stack is a **proposal**. If you strongly prefer another, say so in the group chat before the contracts PR merges. After that it is frozen. Gateway and Market use separate databases on one Postgres instance, with no cross-database joins.

1.2 Golden path (the demo story)

- 1. Ada, a retailer owner, opens the Retailer app. Inventory shows three SKUs running low.

2. In the Console she builds a mandate (payees, per-payment limits, daily and weekly caps) and **signs it in the browser**.
3. The restock agent forecasts demand and commits to pools. The Gateway returns 2 ALLOW (within the auto limit) and 1 ASK (over it).
4. Ada approves the ASK on her phone view. The approval is signed and bound to that exact payment.
5. Neighbours join the pool, the tier price unlocks, the producer is paid and the price difference is refunded.
6. The gullible agent is attacked (poisoned invoice, swapped bank details, x10 quantity, many small payments, fake "already approved"). The Gateway holds. The blast-radius meter shows the cap, not the balance.
7. Audit log verifies, a tamper is caught, the kill switch stops everything.

1.3 Personas and mock auth

Auth is mocked with static tokens from `contracts/seed.json`. Each token works only on its own endpoints.

Token	Who	Works on
<code>tok_owner_ada</code>	Ada, retailer owner	Gateway owner endpoints, Market retailer endpoints, Agent service
<code>tok_producer_prime</code>	Prime Foods Ltd, producer	Market producer endpoints
<code>key_agent_restock</code>	The agent (honest and gullible modes share it)	Gateway agent endpoints, Market read plus orders
<code>key_service_market</code>	Market service	Gateway <code>/internal/*</code>

1.4 Scope

In: everything in section 2 marked P0 or P1. **Out:** real money or bank integrations, KYC, delivery logistics, multi-tenancy, real accounts, native mobile apps, disputes and chargebacks, multi-currency (NGN only), producer onboarding.

Honest labelling. Every screen carries a "MOCK: no real money" badge. In the pitch we say plainly that in production MandatePay instructs a licensed payment partner and never holds money.

1.5 Glossary

Term	Meaning
Principal	The owner whose money is protected (Ada).
Mandate	Owner-signed, immutable rules for one agent: payees, per-payment limits, rolling caps, ask rules.
Payment intent	The agent's <i>proposal</i> to pay. The Gateway decides.
Decision	allow (executes), ask (needs owner's signed approval), block (final).
Approval	Owner's signed OK for one exact intent. Single-use.
Hold / settle	Money parked in escrow for a pool, later released to the producer and/or refunded.
Exposure	The most the agent can still move without a human right now (smallest remaining cap).
Quarantine	After repeated blocks, every otherwise-ALLOW becomes ASK until the owner releases it.

Vocabulary is **AP2-inspired** (Google's Agent Payments Protocol uses intent, cart and payment mandates). Loosely: our mandate is like an intent mandate, our signed approval like a cart mandate, our executed-payment record like a payment mandate. Say "AP2-inspired", never "AP2-compliant", and check the AP2 wording before the pitch.

2. Functional requirements

Priority: **P0** demo-critical, **P1** should have, **P2** if time.

2.1 MandatePay Gateway (Backend)

ID	Requirement	Pri
GW-01	Owner registers an Ed25519 public key. Gateway derives and stores <code>key_id</code> .	P0
GW-02	Create a mandate: schema and semantic validation, signature verification, immutable once stored. A new mandate supersedes the old one atomically.	P0
GW-03	Evaluate payment intents with deterministic ordered rules (4.4) and return allow/ask/block with reasons. No LLM anywhere in the Gateway.	P0
GW-04	Enforce caps atomically: principal row locked, one DB transaction, rolling windows across mandates. Parallel requests cannot overrun a cap.	P0
GW-05	ASK creates an approval. Owner approves with a signature bound to <code>intent_hash</code> . Single-use, with TTL. BLOCK rules re-run at execution.	P0
GW-06	Mock rail: wallet, payee and escrow accounts; transfer, hold, settle; INSUFFICIENT_FUNDS.	P0
GW-07	Hash-chained audit log written in the same transaction as every state change, plus a verify endpoint.	P0
GW-08	Kill switch: while on, every intent is BLOCKed and no approval can execute.	P0
GW-09	Idempotency on payment intents.	P0
GW-10	Role isolation: owner, agent and service tokens each reach only their own endpoints.	P0
GW-11	Auto-quarantine after N blocks in M seconds, with owner release.	P1
GW-12	Revoke a mandate; usage and exposure endpoint for the meter.	P1
GW-13	Demo endpoints: reset and audit tamper (DEMO_MODE only).	P1

2.2 Marketplace (Software developer)

ID	Requirement	Pri
MK-01	Catalog with tiered pricing; seeded producers and SKUs.	P0
MK-02	Retailer inventory: stock levels, movements (sale, receipt, adjustment), sales history.	P0
MK-03	Pools: list, progress to next tier, commit, deadline. Success releases to the producer and refunds the tier difference. Failure refunds in full.	P0
MK-04	Orders (direct and pool commitment) with a Market-computed <code>payment_request</code> . Payment is verified with the Gateway, never on the agent's word.	P0
MK-05	Producer creates a pool and sets tier pricing.	P1
MK-06	Demo helpers: simulate neighbours, close pool now, reset.	P1
MK-07	Pools scoped to a locality (seed one).	P1

2.3 Agent service (ML engineer)

ID	Requirement	Pri
AG-01	Per-SKU demand forecast, reorder point and backtest metrics.	P0
AG-02	Restock recommendations: quantity, urgency, best option (pool or direct) and estimated saving.	P0
AG-03	Restock agent: tool-calling loop that places orders and pays only through <code>request_payment</code> . Readable step trace.	P0
AG-04	Gullible demo agent plus six scripted attack scenarios and running totals.	P0
AG-05	Live-LLM gullible mode (prompt injection through a poisoned invoice) behind a toggle.	P1
AG-06	Simulation: stockout days with versus without the agent.	P2

2.4 Web app (Frontend, Software developer)

ID	Requirement	Pri
UI-01	Persona switcher and MOCK badge on every screen.	P0
UI-02	Console: mandate builder with live JSON preview, in-browser signing, submit.	P0
UI-03	Console: live decision feed (polls audit) with colour-coded allow/ask/block and reason chips.	P0
UI-04	Console: approval inbox. Gateway-rendered summary, visibly untrusted "The agent says" box, hash recomputed before signing, approve or deny.	P0
UI-05	Console: kill switch, always visible.	P0
UI-06	Console: audit view, Verify integrity, demo Tamper button.	P0
UI-07	Console: attack panel and blast-radius meter (balance vs cap, attempts vs breaches).	P0
UI-08	Console: quarantine banner and release, usage meters, one-click Reset.	P1
UI-09	Retail: inventory with days-of-cover.	P0
UI-10	Retail: recommendations and agent run trace.	P0
UI-11	Retail: pool list and detail with progress bar and commit.	P0
UI-12	Retail: orders list. Producer portal to create a pool.	P1

2.5 Pitch (Designer, with the team)

ID	Requirement	Pri
PT-01	Pitch deck (section 9).	P0
PT-02	Timed demo script, 4-minute and 2-minute versions.	P0
PT-03	Backup demo video.	P0
PT-04	At least two real small-lot vs bulk price quotes.	P1
PT-05	Q&A sheet.	P1

3. Non-functional requirements

ID	Area	Requirement	Verified by
NF-01	Isolation	The agent can never edit a mandate, approve, call internal or demo endpoints, or reach the rail. The Gateway makes no LLM calls.	Test: agent token gets 403 on every non-agent endpoint
NF-02	Integrity	A cap is never exceeded under concurrency (principal row lock, one transaction).	<code>scripts/race_test.py</code> : 50 parallel intents, executed total stays at or below the cap
NF-03	Signing	Mandates and approvals verify only against the owner's registered key. Approvals are single-use and bound to <code>intent_hash</code> .	Unit tests on <code>signing.json</code> , replay test, wrong-hash test
NF-04	Auditability	Every state change writes an audit entry in the same transaction. The chain verifies, and tampering is detected.	Verify returns <code>ok: false</code> after tamper
NF-05	Determinism	The policy engine is a pure function with integer math. Same inputs give the same decision.	Table-driven tests, at least one case per reason code
NF-06	Honesty	MOCK badge on every screen. Pitch states the production path plainly.	Review
NF-07	Performance	Intent decision p95 under 300 ms on demo hardware. Pages load under 2 s. Forecast under 1 s per SKU.	Smoke timing
NF-08	Reliability	Reset in under 10 s. Scripted agent mode needs no LLM. Backup video exists.	Rehearsal checklist
NF-09	Usability	Approval takes 3 taps or fewer and works at 360 px width. Console is desktop-first. WCAG AA contrast.	Designer and QA pass
NF-10	Observability	Every request carries a <code>request_id</code> . <code>/healthz</code> on every service. Errors use the common envelope.	Manual
NF-11	Portability	<code>git clone</code> then <code>make dev</code> works on any teammate's laptop. No secrets in git. <code>.env.example</code> is complete.	Fresh-clone test on a second laptop

ID	Area	Requirement	Verified by
NF-12	Contract fidelity	Contracts validate in CI. TypeScript types are generated, never hand-written.	<code>make contracts-check</code>
NF-13	Data	Synthetic data only. No real PII or bank details.	Review

Security invariants the Backend suite must assert: (1) agent token cannot reach owner, internal or demo endpoints; (2) no endpoint modifies a mandate; (3) approval executes only with a valid owner signature over the exact `approval_id`, `intent_id` and `intent_hash`, once; (4) caps hold under concurrency; (5) agent-written text never influences a decision; (6) kill switch blocks approvals too; (7) every state change is audited in its own transaction.

4. Contracts

Files in `/contracts`: `mandate.schema.json`, `gateway.openapi.yaml` (27 operations), `seed.json`, `test-vectors/signing.json`. The Market contract (4.8) becomes `market.openapi.yaml` (Software developer, Wed night). `scripts/check_contracts.py` validates all of it.

4.1 Conventions (all services)

Topic	Rule
Money	Integer minor units (kobo). Never floats. NGN only. <code>{ "amount_minor": 1200000, "currency": "NGN" }</code> = ₦12,000.
Time	UTC, <code>YYYY-MM-DDTHH:MM:SSZ</code> , whole seconds. Browser clocks are not trusted: use <code>server_time</code> from <code>GET /v1/me</code> .
Ids	<code><prefix>_<opaque></code> : <code>prn agt key mdt pay pi apr hld led ord pool sku prd rtl</code> .
JSON	<code>snake_case</code> fields. Enums are lowercase except reason and error codes (UPPER_SNAKE).
Auth	<code>Authorization: Bearer <token></code> . Wrong token kind on an endpoint gives 403.
Errors	<code>{ "error": { "code", "message", "request_id" } }</code> . Codes: UNAUTHENTICATED 401, FORBIDDEN 403, NOT_FOUND 404, CONFLICT 409, IDEMPOTENCY_CONFLICT 409, APPROVAL_NOT_PENDING 409, VALIDATION_ERROR 422, SIGNATURE_INVALID 422, APPROVAL_MISMATCH 422.
Policy outcomes	allow/ask/block are data (HTTP 201 with a body), never HTTP errors.
Idempotency	<code>Idempotency-Key</code> header is required on <code>POST /v1/payment-intents</code> . Market uses the <code>order_id</code> .

4.2 Mandate

Defined by `contracts/mandate.schema.json`. The signed object is the mandate body; the signature travels beside it:

json

```
{ "mandate": { "...body...": "...", "signature": { "alg": "Ed25519", "key_id": "
```

Example body (this exact mandate is signed in `signing.json`):

json

```
{
  "schema_version": "mandatepay/mandate/v1",
  "mandate_id": "mdt_01DEM00000000000000000001",
  "principal_id": "prn_ada",
  "agent_id": "agt_restock",
  "key_id": "key_56475aa75463474c",
  "supersedes": null,
  "issued_at": "2026-10-08T09:00:00Z",
  "valid_from": "2026-10-08T09:00:00Z",
  "valid_until": "2026-10-15T09:00:00Z",
  "currency": "NGN",
  "purpose": "Restock shop inventory from approved suppliers and pools (cap #200,000)",
  "payees": [
    "pay_primefoods",
    "pay_sunbev",
    "pay_market_escrow"
  ],
  "limits": {
    "per_txn": {
      "auto_max_minor": 5000000,
      "hard_max_minor": 15000000
    },
    "windows": [
      {
        "name": "daily",
        "seconds": 86400,
        "cap_minor": 20000000
      },
      {
        "name": "weekly",
        "seconds": 604800,
        "cap_minor": 60000000
      }
    ]
  },
  "ask_rules": {
    "approval_ttl_seconds": 900,
    "anomaly": {
      "enabled": true,
      "history_count": 5,
      "percent_of_average": 300
    }
  },
  "quarantine": {
    "blocked_attempts": 3,
    "window_seconds": 600
  }
}
```

```
}
}
```

Server-side rules that JSON Schema cannot express (Backend implements; Frontend mirrors them in form validation):

1. `auto_max_minor <= hard_max_minor`.
2. `valid_until > valid_from`, span at most 30 days, `valid_from >= issued_at`.
3. `issued_at` within 300 s of server time.
4. `principal_id` equals the authenticated owner. `agent_id` belongs to that principal. `key_id` is registered to that principal.
5. Every id in `payees` exists in the registry.
6. Window names are unique.
7. `mandate_id` not already used, otherwise 409.
8. If the agent already has an active mandate, `supersedes` must name it, otherwise 409. Acceptance supersedes the old one in the same transaction.

Loosening needs a signature, tightening does not. New or larger authority (mandate, approval) is signed. Reducing authority (kill switch, revoke, deny, quarantine) only needs the owner session.

4.3 Signing and hashing

Item	Rule
Canonical bytes	RFC 8785 (JCS), UTF-8. Signed objects have no floats , so <code>json.dumps(obj, sort_keys=True, separators=(",", ":"), ensure_ascii=False)</code> is byte-identical.
Signature	Ed25519 over the canonical bytes. Encoded base64url, no padding (86 chars).
Public key	Raw 32 bytes, base64url, no padding (43 chars).
<code>key_id</code>	<code>"key_"</code> + first 16 hex chars of SHA-256(raw public key). The client can compute it before registering.
<code>content_hash</code>	Lowercase hex SHA-256 of the canonical mandate body.
<code>intent_hash</code>	SHA-256 of canonical JSON of the bound fields : <code>{ mandate_id, payee_id, amount_minor, currency, reference, expires_at }</code> . <code>payee_id</code> is the payee the Gateway <i>resolved</i> , not the agent's claim.
Approval statement	<code>{ "type": "mandatepay/approval/v1", "approval_id", "intent_id", "intent_hash", "decision": "approve" }</code> , signed with the same rules. No client timestamp, so clock skew cannot break approvals.

What you see is what you sign. The approval screen shows the `bound` values, recomputes `intent_hash` from them, and refuses to sign if it differs from the Gateway's `intent_hash`.

Libraries. Python: `rfc8785`, `cryptography`. TypeScript: `canonicalize`, `@noble/ed25519` (or WebCrypto Ed25519). `contracts/test-vectors/signing.json` holds a demo key, a signed mandate, an `intent_hash` and a signed approval. Both Backend and Frontend must reproduce **every** derived value from the inputs. The vector was verified in Python (`rfc8785` + `cryptography`) and in Node (`canonicalize` + built-in Ed25519) before publishing.

4.4 Decision engine

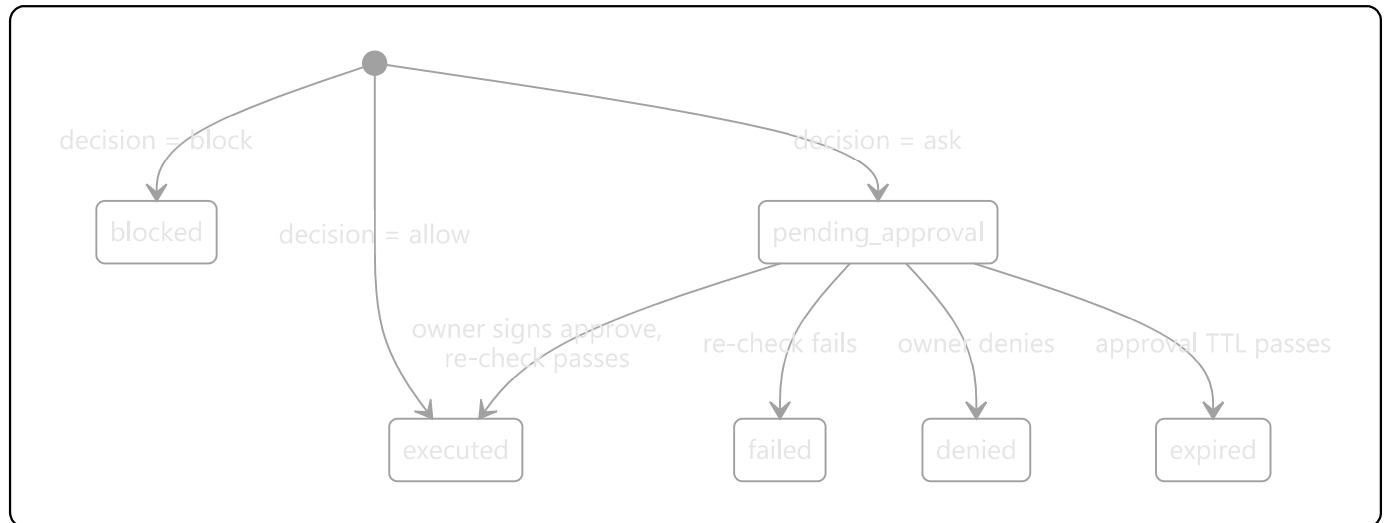
Pure function in `services/gateway/app/policy.py`: no I/O, integer math only. Before evaluation, a repeated `Idempotency-Key` returns the stored result. Rules run in this fixed order. **The first BLOCK ends evaluation.** If nothing blocks, every ASK rule is evaluated and all hits are returned (primary reason first).

#	Code	Result	Fires when
1	KILL_SWITCH_ACTIVE	BLOCK	Owner's kill switch is on.
2	NO_ACTIVE_MANDATE, MANDATE_NOT_YET_VALID, MANDATE_EXPIRED	BLOCK	No active mandate for this agent, or now is outside <code>valid_from</code> to <code>valid_until</code> .
3	CURRENCY_MISMATCH	BLOCK	Intent currency differs from the mandate's.
4	PAYEE_UNKNOWN	BLOCK	Only <code>payee_id</code> given and not in the registry; or only <code>destination</code> given and no payee has that bank code and account number.
4	DESTINATION_MISMATCH	BLOCK	Both given, and the destination differs from the registered destination of that <code>payee_id</code> (bank code or account number).
4	PAYEE_NOT_ALLOWED	BLOCK	Resolved payee is not in the mandate's <code>payees</code> .
5	INTENT_EXPIRED	BLOCK	<code>expires_at</code> is in the past. Default <code>expires_at</code> is <code>created_at</code> + 900 s.
6	PER_TXN_HARD_MAX_EXCEEDED	BLOCK	Amount above <code>hard_max_minor</code> .
7	WINDOW_CAP_EXCEEDED	BLOCK	For any window, spend + amount would exceed <code>cap_minor</code> . Spend = executed payments by this agent for this principal in the last <code>seconds</code> , across all mandates. Refunds do not restore headroom (conservative). Equal to the cap is allowed.
8	INSUFFICIENT_FUNDS	BLOCK	Wallet balance is below the amount.
9	QUARANTINED	ASK	Mandate is quarantined.
10	OVER_AUTO_LIMIT	ASK	Amount above <code>auto_max_minor</code> (and at or below <code>hard_max_minor</code>).
11	ANOMALOUS_AMOUNT	ASK	Anomaly rule enabled, at least <code>history_count</code> executed payments to this payee exist, and $\frac{\text{amount} * 100 * n}{\text{percent_of_average} * \text{sum}}$ over the last <code>n = history_count</code> of them (integer math).

Atomicity: every transaction that decides or executes an intent first locks the principal row (`(SELECT ... FOR UPDATE)`), then reads state, decides, writes the intent, ledger and audit rows, and commits. Approvals execute through the same path.

Quarantine: count BLOCK decisions for this agent after the later of (now minus `window_seconds`, last release). On reaching `blocked_attempts`, set `quarantined`, log `quarantine.entered`. The triggering intent stays BLOCK.

4.5 Intent and approval lifecycle



- ASK creates an approval with `expires_at = min(now + approval_ttl_seconds, intent.expires_at)`.
- `POST /v1/approvals/{id}/decision` verifies the signature, that the statement matches the stored `approval_id`, `intent_id` and `intent_hash`, and that the approval is `pending`. Then, under the principal lock, it **re-runs rules 1 to 8** (not the ASK rules) and executes. Caps cannot be approved away.
- A signed approval is single-use: consumed even if the re-check fails (intent becomes `failed` with a `failure` reason).
- Payments to the escrow payee create a `hold` ledger entry and return a `hold_id`.

4.6 Audit log

Single hash chain. `seq` starts at 1 and has no gaps.

`hash = SHA-256( prev_hash + canonical_json({ seq, ts, type, principal_id, actor, subject, data }) )`, where `prev_hash` is the previous 64-char hex string (genesis is 64 zeros) concatenated with the canonical JSON text, hashed as UTF-8. Call it **tamper-evident, not tamper-proof**.

Event type	data
key.registered	{ key_id }
mandate.created / mandate.superseded / mandate.revoked	{ mandate_id, content_hash, supersedes } / { mandate_id, superseded_by } / { mandate_id }
intent.decided	{ intent_id, decision, status, reason_codes[], amount_minor, currency, payee_id, reference }
approval.requested / approval.resolved	{ approval_id, intent_id, expires_at } / { approval_id, intent_id, outcome, intent_status }
payment.executed / payment.failed	{ intent_id, ledger_entry_id, hold_id } / { intent_id, code }
hold.settled	{ hold_id, released_minor, refunded_minor, payee_id }
kill_switch.changed	{ active, reason }
quarantine.entered / quarantine.released	{ mandate_id, blocked_count } / { mandate_id }
request.rejected	{ endpoint, http_status, error_code } (malformed or unauthorised agent calls)

The console live feed polls `GET /v1/audit?after_seq=N` every second. SSE is a stretch goal, not part of the contract.

4.7 Gateway API

Full schemas, examples and error shapes: `contracts/gateway.openapi.yaml`. Generated summary:

Method	Path	Caller	operationId	Purpose
GET	/healthz	none	healthz	Liveness probe (no auth)
GET	/v1/me	owner	getMe	Current owner, registered keys, kill-switch state and server time
POST	/v1/keys	owner	registerKey	Register the owner's Ed25519 public key (generated in the browser)
GET	/v1/payees	owner	listPayees	Registered payees (for the mandate builder's allowlist picker)
GET	/v1/mandates	owner	listMandates	List mandates, newest first
POST	/v1/mandates	owner	createMandate	Submit a signed mandate
GET	/v1/mandates/{mandate_id}	owner	getMandate	Get one mandate
POST	/v1/mandates/{mandate_id}/revoke	owner	revokeMandate	Revoke a mandate (tightening needs no signature)
GET	/v1/mandates/{mandate_id}/usage	owner	getMandateUsage	Rolling-window spend, headroom and exposure (feeds the blast-radius meter)
POST	/v1/mandates/{mandate_id}/release-quarantine	owner	releaseQuarantine	Owner lifts quarantine (blocks before this moment stop counting)
GET	/v1/agent/mandate	agent	getAgentMandate	The agent's own active mandate

Method	Path	Caller	operationId	Purpose
				and current usage (read-only)
POST	/v1/payment-intents	agent	createPaymentIntent	Propose a payment. Returns the decision (allow / ask / block) as data.
GET	/v1/payment-intents	owner	listPaymentIntents	List intents, newest first
GET	/v1/payment-intents/{intent_id}	agent, owner	getPaymentIntent	Get one intent (agent: its own; owner: any of theirs). Poll this after an ASK.
GET	/v1/approvals	owner	listApprovals	Approval inbox
GET	/v1/approvals/{approval_id}	owner	getApproval	One approval, with the gateway-rendered summary to show the owner
POST	/v1/approvals/{approval_id}/decision	owner	decideApproval	Approve (signed) or deny
GET	/v1/kill-switch	owner	getKillSwitch	Kill-switch state
PUT	/v1/kill-switch	owner	setKillSwitch	Turn the kill switch on or off. While on, every intent is BLOCKed and approvals cannot execute.
GET	/v1/audit	owner	listAudit	Hash-chained audit log, oldest first. The console polls this with after_seq for the live feed.

Method	Path	Caller	operationId	Purpose
GET	/v1/audit/verify	owner	verifyAudit	Recompute the whole hash chain server-side ("Verify integrity" button)
GET	/v1/accounts	owner	listAccounts	Mock rail balances (owner wallet, escrow, payee accounts)
GET	/v1/ledger	owner	listLedger	Mock rail ledger entries, newest first
GET	/internal/v1/intents/{intent_id}	service	getIntentInternal	Market verifies a payment itself instead of trusting the agent's word
POST	/internal/v1/holds/{hold_id}/settle	service	settleHold	Settle an escrow hold atomically - release to a producer and/or refund the originator
POST	/demo/v1/reset	owner	resetDemo	Wipe and reseed from contracts/seed.json (DEMO_MODE only). Owner must re-register key and re-sign a mandate.
POST	/demo/v1/tamper-audit	owner	tamperAudit	Edit one audit row directly in the DB so "Verify integrity" fails on stage (DEMO_MODE only)

Notes:

- `Agent` endpoints are the only ones the agent token can call. `Internal` is for the Market service only. `Demo` exists only when `DEMO_MODE=true`.
- Mock it before the real thing exists: `npx @stoplight/prism-cli mock contracts/gateway.openapi.yaml -p 4010`, then send `Prefer: example=ask` (or `allow`, `block`) on `POST /v1/payment-intents`.
- Frontend generates types with `openapi-typescript`. Nobody hand-writes API types.

4.8 Market API (v0, Software developer formalises as `market.openapi.yaml` by Wed night)

Same conventions as 4.1. Scoping comes from the token: the retailer is "the caller's retailer". Additive changes only after the lock.

Method	Path	Caller	Purpose
GET	/healthz	none	Liveness
GET	/v1/products	any	Catalog with tier pricing
GET	/v1/inventory	owner, agent	Stock for the caller's retailer
POST	/v1/inventory/{sku_id}/movements	owner	Record <code>sale</code> , <code>receipt</code> or <code>adjustment</code>
GET	/v1/sales?sku_id=&days=	owner, agent	Daily sales series
GET	/v1/pools?status=&sku_id=	owner, agent, producer	Pools with progress
GET	/v1/pools/{pool_id}	any	One pool
POST	/v1/pools	producer	Create a pool (P1)
POST	/v1/orders	owner, agent	Create an order draft; returns <code>payment_request</code>
GET	/v1/orders, /v1/orders/{order_id}	owner, agent	List and read orders
POST	/v1/orders/{order_id}/payment	owner, agent	Attach <code>{ intent_id }</code> ; Market verifies with the Gateway
POST	/demo/v1/pools/{pool_id}/simulate-neighbors	owner	Body <code>{ qty }</code> . Simulated commitments never touch the rail.
POST	/demo/v1/pools/{pool_id}/close	owner	Force the deadline and settle now
POST	/demo/v1/reset	owner	Reseed from <code>seed.json</code> plus <code>data/catalog.json</code>

json

```
// Pool
{ "pool_id": "pool_noodles_1", "sku_id": "sku_noodles_carton", "sku_name": "Instant
  "producer_id": "prd_primefoods", "locality": "demo-locality", "status": "open",
  "moq": 50, "committed_qty": 32,
  "tiers": [ { "min_qty": 1, "unit_price_minor": 1200000 }, { "min_qty": 50, "unit
    "current_unit_price_minor": 1200000, "next_tier": { "min_qty": 50, "unit_price_m
// status: open, closed_success, closed_failed. progress_pct = committed_qty towar

// Order (response of POST /v1/orders with { kind, sku_id, qty, pool_id? })
{ "order_id": "ord_01HX...", "kind": "pool_commitment", "sku_id": "sku_noodles_car
  "status": "awaiting_payment", "amount": { "amount_minor": 12000000, "currency":
  "payment_request": { "payee_id": "pay_market_escrow", "amount": { "amount_minor"
    "reference": "ord_01HX...", "description": "Pool commitment
  "intent_id": null, "created_at": "2026-10-08T10:00:00Z" }
// kind: pool_commitment, direct. status: awaiting_payment, awaiting_approval, pai
```

Rules:

- $\text{amount} = \text{qty} \times \text{tiers}[0].\text{unit_price_minor}$ (smallest-lot price). Pool commitments pay `pay_market_escrow`; direct orders pay the producer's payee.
- **The agent never computes money.** `payment_request` comes from Market. The honest agent's `request_payment` sends exactly that.
- Reconcile loop (every 2 s) for orders with an `intent_id`: read `GET /internal/v1/intents/{id}` from the Gateway. Mark `paid` **only if** status is `executed` AND payee, amount and reference match the order's `payment_request`. `pending_approval` gives `awaiting_approval`. blocked, denied, expired or failed gives `blocked`. A mismatch is never `paid`.
- Pool close (deadline or demo close): success if $\text{committed_qty} \geq \text{moq}$. Final price = tier with the highest `min_qty` at or below `committed_qty`. For each paid real commitment call `settle(hold_id, release_to_payee_id = producer payee, release_minor = qty x final_price, refund_minor = hold - release)`. Failure: `release_minor = 0`, refund all.

4.9 Agent service API (ML engineer)

Callers use `tok_owner_ada`. Same conventions as 4.1.

Method	Path	Purpose
GET	<code>/agent/v1/forecast?sku_id=</code>	Forecast, reorder point, backtest
GET	<code>/agent/v1/recommendations</code>	Restock recommendations for the caller's retailer
POST	<code>/agent/v1/restock/runs</code>	Body <code>{ mode: "scripted" or "llm" }</code> . Starts a run, returns <code>{ run_id, status }</code>
GET	<code>/agent/v1/restock/runs/{run_id}</code>	Step trace and outcomes
GET	<code>/agent/v1/attacks</code>	Scenario list (id, title, description, expected)
POST	<code>/agent/v1/attacks/{scenario_id}/run</code>	Run one scenario (<code>S1</code> to <code>S6</code>)
GET	<code>/agent/v1/attacks/summary</code>	Running totals for the blast-radius meter
POST	<code>/demo/v1/reset</code>	Clear run and attack history

json

```
// Forecast
{ "sku_id": "sku_noodles_carton", "horizon_days": 14, "daily_forecast": [ { "date"
  "lead_time_days": 2, "safety_stock": 6, "reorder_point": 20, "on_hand": 14, "day
  "backtest": { "method": "holt_winters", "window_days": 14, "mae": 1.4, "wape": 0
// Recommendation
{ "sku_id": "sku_noodles_carton", "urgency": "now", "recommended_qty": 30, "reason
  "best_option": { "type": "pool", "pool_id": "pool_noodles_1", "est_unit_price_mi
// Attack run result
{ "scenario_id": "S3", "attempts": [ { "intent_id": "pi_...", "decision": "ask", "
  "executed_total_minor": 0, "verdict": "contained" }
// Summary
{ "attempts": 6, "blocked": 2, "asked": 2, "executed": 0, "executed_total_minor":
```

`verdict` is `contained` unless money moved to a non-allowlisted payee or total executed exceeded a cap (`breaches` counts those; the target is 0).

Agent tool names (fixed, shared by Agent and Frontend trace view): `get_inventory`, `get_forecast`, `list_pools`, `get_mandate`, `create_order`, `request_payment`, `get_payment_status`.

- Honest `request_payment(order_id)` builds the intent only from the order's `payment_request` (payee, amount, reference) with `Idempotency-Key = order_id`, then calls `POST /v1/orders/{id}/payment`.

- The gullible agent also has `request_payment_raw(payee_id, destination, amount, reference, description)`. It exists only to simulate a fooled agent. The Gateway must hold even then.
- System-prompt rule for the live agent: if a payment is blocked, stop and report to the owner. Never retry with a different payee.

4.10 Shared seed and demo constants

All seed scripts read `contracts/seed.json`. Key numbers (demo numbers, so changing them means updating section 8):

Item	Value
Ada's wallet	₦2,000,000
Default mandate	auto limit ₦50,000, hard max ₦150,000, daily cap ₦200,000, weekly cap ₦600,000, approval TTL 15 min, quarantine at 3 blocks in 10 min
Anomaly rule	ASK above 300% of the average of the last 5 payments to that payee
History	5 past payments of ₦12,000 to <code>pay_primefoods</code> , 8 to 30 days old (never counts toward caps)
Payees	<code>pay_primefoods</code> , <code>pay_sunbev</code> , <code>pay_market_escrow</code> (all in the default mandate); <code>pay_greenfarms</code> (registered, not allowlisted)
Attacker	Unregistered account <code>9999999999</code> "Urgent Settlement Services"; swapped Prime Foods account <code>1001999999</code>
Golden path constraint	Catalog prices must make the restock run produce 2 ALLOW and 1 ASK under the default mandate

5. Repo and working agreement

```
.
├─ README.md  Makefile  docker-compose.yml  .env.example
├─ .github/   CODEOWNERS  pull_request_template.md  workflows/ci.yml
├─ contracts/                                     # SOURCE OF TRUTH, changed by PR only
│   └─ mandate.schema.json  gateway.openapi.yaml  market.openapi.yaml
seed.json
├─ test-vectors/signing.json
├─ docs/      PROJECT_SPEC.md  DEMO_SCRIPT.md  pitch/ (deck PDF, assets/)
├─ design/    tokens.json  exports/
├─ data/      catalog.json  sales_90d.csv
├─ scripts/   check_contracts.py  gen_sales.py  race_test.py  e2e.sh
reset.sh
├─ services/  gateway/ (Backend)  market/ (Software dev)  agent/ (ML)
├─ apps/web/  app/(console)/ (Frontend)  app/(retail)/ (Software dev)
components/ui-kit/ (Frontend)
```

Commands (Software developer creates them tonight): `make dev` (Postgres via Docker, three services and web with reload), `make mocks` (Prism on 4010 and 4011), `make seed`, `make reset` (calls all three `/demo/v1/reset`), `make test`, `make contracts-check`.

Env vars (`.env.example` lists all): `GATEWAY_URL`, `MARKET_URL`, `AGENT_URL` (defaults `http://localhost:8001/8002/8003`), `GATEWAY_DATABASE_URL`, `MARKET_DATABASE_URL`, `DEMO_MODE=true`, `LLM_API_KEY` (your local `.env` or host secrets only), and `NEXT_PUBLIC_GATEWAY_URL`, `NEXT_PUBLIC_MARKET_URL`, `NEXT_PUBLIC_AGENT_URL` for the web app.

Git rules

1. Branch from `main` as `feat/<area>-<what>`, for example `feat/gateway-policy-engine`.
One task, one branch, one PR where possible.
2. `git pull --rebase origin main` before pushing. No force-push to `main`. Squash-merge.
3. CI must be green. Review is required only for `contracts/`, `docker-compose.yml` and `Makefile`; everything else merges when green, then post in the group chat.
4. Touch only your own directories. Need a change elsewhere? Send a small PR to the owner or ask in chat. Shared root files belong to the Software developer.
5. Contract changes: PR labelled `contract-change`, approved by the endpoint's owner and at least one consumer, announced in chat. After the lock, additive only.
6. No secrets in git. Commit small, imperative messages ("Add window cap rule").

```
# .github/CODEOWNERS (fill in real handles)
/contracts/ @backend @software-dev
/services/gateway/ @backend
/services/market/ @software-dev
/services/agent/ @ml
/apps/web/app/(console)/ @frontend
/apps/web/components/ui-kit/ @frontend
/apps/web/app/(retail)/ @software-dev
/design/ /docs/pitch/ @designer
/Makefile /docker-compose.yml /.github/ @software-dev
```

6. Tasks per person

Format: **ID • due • title**. Then what to build, and **Done when** (a check anyone can run). Times are proposals; shift them to the real deadline, but keep the order.

6.1 Backend: `services/gateway` (the heart)

Needs from others: nothing to start. Gives Frontend the signing rules and endpoints, Market the internal API.

❑ **B1 • Wed 21:00 • Scaffold, auth, seed.** FastAPI app, Alembic migrations (principals, agents, owner_keys, payees, mandates, payment_intents, approvals, accounts, ledger_entries, audit_log, idempotency_keys, kill_switches). Bearer middleware for the three token kinds from `seed.json`. `GET /healthz`, `GET /v1/me`, `POST /demo/v1/reset` (wipe and reseed, including the 5 history payments). **Done when:** `GET /v1/me` works with `tok_owner_ada`, the agent key gets 403 on it, and `make seed` is repeatable.

❑ **B2 • Wed 22:00 • Crypto module.** `app/crypto.py`: canonical JSON (`rfc8785`), SHA-256, Ed25519 verify, `key_id` derivation, `intent_hash`. **Done when:** `tests/test_signing.py` reproduces every value in `signing.json` and rejects a one-byte change.

❑ **B3 • Wed 23:30 • Keys and mandates.** `POST /v1/keys` (idempotent), `GET /v1/payees`, `POST /v1/mandates` (schema, rules 4.2, signature, supersede in one transaction), `GET /v1/mandates`, `GET /v1/mandates/{id}`, `revoke`, `usage`. **Done when:** the vector mandate is accepted, a second mandate supersedes it, an edited payload returns 422 `SIGNATURE_INVALID`. Unblocks F2 and F3.

❑ **B4 • Thu 10:00 • Policy engine.** `app/policy.py` pure function per 4.4. `tests/policy_cases.json`: at least one case per reason code plus the six scenarios in section 8. **Done when:** all cases pass.

❑ **B5 • Thu 11:30 (M1) • Intents and mock rail.** `POST /v1/payment-intents` with idempotency, principal-row `FOR UPDATE`, decide, persist, rail transfer or hold, audit, all in one transaction. `GET` intents, `/v1/agent/mandate`, `/v1/accounts`, `/v1/ledger`. **Done when:** allow, ask and block each return the right status; a repeated key replays with 200; low balance gives `INSUFFICIENT_FUNDS`.

- ❑ **B6 · Thu 13:30 · Approvals.** ASK creates the approval; `GET /v1/approvals`, `GET /v1/approvals/{id}`, `POST .../decision` per 4.5. **Done when:** wrong key gives 422, changed hash gives `APPROVAL_MISMATCH`, a second approve gives 409, and a cap exhausted between ask and approve leaves the intent `failed`.
- ❑ **B7 · Thu 15:00 · Audit chain, kill switch, quarantine.** Chain per 4.6, `GET /v1/audit`, `GET /v1/audit/verify`, `POST /demo/v1/tamper-audit`, `PUT /v1/kill-switch`, quarantine counter and release. **Done when:** tamper makes verify return `ok: false` with `first_bad_seq`; the kill switch blocks new intents and approvals; 3 blocks in 10 minutes enters quarantine and the next would-be ALLOW becomes ASK `QUARANTINED`.
- ❑ **B8 · Thu 15:30 (M2) · Internal API for Market.** `GET /internal/v1/intents/{id}`, `POST /internal/v1/holds/{hold_id}/settle` (atomic, amounts must sum to the hold). **Done when:** Market's pool settlement runs end to end.
- ❑ **B9 · Thu 16:30 · Race and invariant tests.** `scripts/race_test.py`: 50 parallel ~~¥~~40,000 intents; executed total must not exceed ¥200,000. Tests for the security invariants in section 3. **Done when:** green 10 runs in a row.
- ❑ **B10 · Thu 17:00 onward · Support.** Fix what the attack runs expose, keep reset under 10 s, pair with Frontend on any signing issue.

6.2 Software developer: repo, deploy, `services/market`, retailer app

Needs: B-contracts (already here), ML's `gen_sales.py` (Thu 10:00), Designer's quotes for prices.

- ❑ **S1 · Wed 20:30 · Repo scaffold (do this first, everyone is waiting).** Monorepo per section 5, drop in this kit's files, `docker-compose.yml` (Postgres 16 with two databases), Makefile, `.env.example`, CODEOWNERS, PR template, GitHub Actions (lint plus `make contracts-check`), README quickstart. Each service ships a stub with `/healthz`. **Done when:** a fresh clone plus `make dev` brings up all four stubs.
- ❑ **S2 · Wed 22:00 · Mock servers.** `make mocks` runs Prism on the gateway and market OpenAPI files. **Done when:** Frontend and ML are calling them.
- ❑ **S3 · Wed 22:00 · `contracts/market.openapi.yaml`.** Turn 4.8 into OpenAPI 3.1 with the same conventions and examples for Pool and Order. PR reviewed by ML and Frontend. **Done when:** merged and `make contracts-check` passes. This is the contract lock for your side.
- ❑ **S4 · Thu 09:00 · Deploy skeleton.** Pick a host (for example Vercel for web, Render, Railway or Fly for services plus Postgres). Public URL for each `/healthz`, secrets in host settings. **Done when:** URLs are in the README and a push to `main` redeploys.
- ❑ **S5 · Thu 10:00 · Market data and seed.** Tables, then seed from `contracts/seed.json` plus `data/catalog.json` (6 SKUs, 3 producers, tiers, one open pool per SKU, one locality). Prices from the Designer's real quotes, marked illustrative until they arrive. Sales from ML's

`data/sales_90d.csv`). **Done when:** the golden-path constraint in 4.10 holds (2 ALLOW and 1 ASK).

❑ **S6 • Thu 12:00 (M1) • Market API.** Everything in 4.8 except pool settlement: products, inventory and movements, sales, pools, orders with `payment_request`, `POST /v1/orders/{id}/payment`, the 2-second reconcile loop. **Done when:** order, agent pays, status becomes `paid`; a payment with the wrong amount or payee never becomes `paid`.

❑ **S7 • Thu 14:30 • Pool lifecycle.** Deadline job, `simulate-neighbors`, `close`, settlement through the Gateway `settle` call (success: release `qty x final price` and refund the difference; failure: refund all). **Done when:** both paths show correct ledger entries in the console.

❑ **S8 • Thu 12:00 to 18:00 • Retailer app** (`apps/web/app/(retail)`). Inventory (days-of-cover badge, record a sale or receipt), Recommendations with agent run trace, Pools (progress bar, commit), then Orders and Producer portal (P1). Use Frontend's UI kit and the Designer's tokens. **Done when:** the golden path runs from the UI alone.

❑ **S9 • Thu 18:30 • End-to-end and reset.** `scripts/e2e.sh` (golden path against any base URL) and `scripts/reset.sh` (all three resets). **Done when:** reset plus e2e is green on the **deployed** URLs.

6.3 ML engineer: `services/agent`

Needs: SD's `data/catalog.json` for SKU ids (agree tonight), Gateway mock for client work.

❑ **M1 • Wed 21:00 • Sales generator.** `scripts/gen_sales.py` writes `data/sales_90d.csv` (`(date, sku_id, qty)`) for 6 SKUs: fixed random seed, weekly seasonality, month-end uplift, noise. **Done when:** SD loads it into the Market seed.

❑ **M2 • Wed 23:30 • Forecast and backtest.** `forecast.py`: seasonal-naive baseline versus Holt-Winters (or another simple, explainable model). Backtest on the last 14 days: MAE, WAPE, baseline WAPE. Reorder point = lead-time demand + safety stock ($z = 1.65 \times \text{residual sigma} \times \sqrt{\text{lead time}}$). `GET /agent/v1/forecast`. **Done when:** every SKU returns the 4.9 shape, and the report is honest about SKUs where the model does not beat the baseline.

❑ **M3 • Wed night • Clients.** Thin typed wrappers for Gateway and Market in `clients/`, tested against the Prism mocks. **Done when:** `request_payment` runs against the gateway mock with `Prefer: example=ask`.

❑ **M4 • Thu 11:30 (M1) • Recommendations and restock agent.** `GET /agent/v1/recommendations`; `POST /agent/v1/restock/runs` with `scripted` (deterministic planner, no LLM, the demo default) and `llm` modes; the tool names and rules in 4.9; step trace. **Done when:** a run on seed data places 3 orders and the Gateway returns 2 ALLOW and 1 ASK, with a readable trace.

❑ **M5 • Thu 15:00 • Gullible agent and attack runner.** Six scenarios per section 8 in `attacks/scenarios/`, `request_payment_raw`, the three attack endpoints. S4 and S6 assert

invariants (executed total at or below the cap), not hard-coded counts. **Done when:** every scenario matches its expected decision and reason codes on a fresh reset.

❑ **M6 • Thu 17:30 • Evidence for the deck.** Backtest chart (forecast vs actual) and, if time, a 60-day stockout simulation with and without the agent (AG-06). PNG or SVG into `(docs/pitch/assets/)`. **Done when:** the Designer has the files.

❑ **M7 • Thu PM, only after M5 is green • Live-LLM mode (bonus).** Feed a poisoned invoice to the real LLM and show it attempting the bad payment. Toggle in the attack panel; fall back to scripted on any LLM error.

6.4 Frontend: `(apps/web/app/(console))` and `(components/ui-kit)`

Needs: Backend's B2 and B3 for real signing; the Prism mock until then.

❑ **F1 • Wed 21:30 • Scaffold.** Next.js App Router, TypeScript, Tailwind, shadcn/ui, route groups `(console)` and `(retail)`. `(npm run gen:api)` runs `(openapi-typescript)` on the gateway file (market file once S3 lands). Typed fetch client. Persona switcher using the static tokens. MOCK badge component. **Done when:** the console shell renders against the Prism mock.

❑ **F2 • Wed 23:30 • In-browser signing.** `(lib/crypto.ts)`: generate and store an Ed25519 key (IndexedDB), derive `(key_id)`, canonicalize, sign, register the key on first run **and after every reset**. Vitest reproduces `(signing.json)`. **Done when:** the test passes. Signing is the product: do not fake it (see section 10).

❑ **F3 • Thu 10:30 • Mandate builder.** Form (payees from `(GET /v1/payees)`, limits, windows, anomaly, TTL), a "Use default demo mandate" button from `(seed.json)`, live JSON preview, `(server_time)` from `(/v1/me)` for timestamps, sign, `(POST /v1/mandates)`, show status and usage meters. **Done when:** the real Gateway accepts a mandate signed in your browser.

❑ **F4 • Thu 12:00 (M1) • Live decision feed.** Poll `(GET /v1/audit?after_seq=)` every second. Rows colour-coded allow/ask/block with reason chips and amounts. **Done when:** an agent run shows up within 2 seconds.

❑ **F5 • Thu 13:30 • Approval inbox (mobile-first).** Pending list and detail from `(GET /v1/approvals)`; show the gateway `(summary)` and `(bound)`; the "The agent says" box visibly marked untrusted; recompute `(intent_hash)` from `(bound)` and refuse to sign on mismatch; approve (sign) or deny; expiry countdown. **Done when:** an ASK is approved end to end and a tampered hash is refused.

❑ **F6 • Thu 15:00 • Kill switch, quarantine, audit.** Big always-visible kill switch, quarantine banner with release, audit view with Verify integrity (green or red) and the demo Tamper button. **Done when:** tamper then verify turns red.

❑ **F7 • Thu 16:00 (M2) • Attack panel and blast-radius meter.** Scenarios from `(GET /agent/v1/attacks)`, Run and Run all, results table. Meter from `(GET /v1/mandates/{id}/usage)`, `(GET /v1/accounts)` and the attack summary: balance versus cap, attempts versus breaches, executed total. One-click Reset (three resets, then re-sign the default mandate). **Done when:** all six scenarios run from the UI.

☐ **F8 • Thu 17:30 • Polish.** Apply the Designer's tokens and hero screens, empty, loading and error states, responsive pass, copy. Then help Software dev with shared components (tables, progress bar).

6.5 Designer: design system, pitch deck, demo

Needs: nothing to start. Gives tokens and screens to Frontend, real prices to Software dev.

☐ **D1 • Wed night (1 hour) • Real quotes.** Call or WhatsApp a wholesaler or shopkeeper and get 2 or 3 small-lot versus bulk prices for real products: name, date, SKU, unit, price, notes. Send prices to Software dev. If nobody answers by Thu 10:00, use placeholders and label them "illustrative". **Done when:** the numbers are in `docs/pitch/quotes.md`.

☐ **D2 • Wed night • Name and identity.** Three platform name options and a MandatePay wordmark. Team votes by Thu 09:00.

☐ **D3 • Thu 09:00 • Tokens and 4 hero screens.** `design/tokens.json` (colours including semantic allow, ask and block, type scale, spacing, radius) plus a Tailwind snippet. Figma frames: (1) approval card, mobile; (2) console live feed with blast-radius meter; (3) mandate builder and sign; (4) pool detail with progress bar and agent recommendation. PNGs into `design/exports/`. **Done when:** Frontend confirms they can build from it.

☐ **D4 • Thu 12:00 • Deck v1.** Storyline and placeholders per section 9. **D5 • Thu 18:00 • Deck v2** with real screenshots and numbers (quotes, backtest, attack results). **D6 • Thu 22:00 • Deck final,** PDF plus editable link in `docs/pitch/`.

☐ **D7 • Thu 16:00 • Demo script.** `docs/DEMO_SCRIPT.md`: exact clicks, who drives, who speaks, 4-minute and 2-minute versions (see section 8).

☐ **D8 • Thu night • Backup video.** Record the demo from the **deployed** URL with voice-over. **Done when:** the file is on two laptops and a phone.

☐ **D9 • Thu night to Fri • Q&A sheet and rehearsals.** Draft the answers in 9.3, then run three timed rehearsals on Friday.

7. Timeline and checkpoints

When	Milestone	Done means
Wed 22:00	M0 Foundations and contract lock	Repo live (S1), market contract merged (S3), mocks running (S2), B1 and B2, F1 and M1 done
Thu 09:00	Design ready	Tokens and 4 hero screens (D3), platform name chosen, deploy skeleton (S4), B3 done
Thu 12:00	M1 Ugly end-to-end	Agent run, then order, then intent, then ALLOW, then order (paid), visible in the console feed (B5, S6, M4, F4)
Thu 16:00	M2 Full demo path	ASK then approve; all six attacks from the console; kill switch; audit verify; pool settlement (B6 to B8, M5, F5 to F7, S7)
Thu 19:00	M3 Feature freeze	Bug fixes, copy and polish only. Deck v2 done. e2e green on the deployed URLs
Thu 22:00	M4 Demo locked	Deck final, backup video recorded, deploy frozen, reset plus e2e verified on deployed URLs
Fri	Rehearse and pitch	Three timed run-throughs, P0 bug fixes only, offline copies of everything

Stand-ups: 10 minutes at Thu 09:30, 12:00, 16:00 and 19:00. If you are blocked for more than 20 minutes, post in the group chat. If a contract is the problem, fix the contract by PR rather than working around it.

8. Demo plan and attack scenarios

8.1 Golden path (4-minute version)

Time	Step	Shows
0:00	Ada opens the Retailer app. Inventory shows 3 SKUs low.	The retailer's pain
0:30	Console: "Use default demo mandate", sign in the browser.	Owner-signed rules
1:00	Retail: Run restock agent. Trace: forecast, 3 pool commitments. Gateway: 2 ALLOW, 1 ASK.	The agent working inside rules
1:45	Phone view: approval card, "The agent says" box, Approve (sign).	Human in the loop, signed and bound
2:15	Simulate neighbours, close pool. Tier price unlocks, refund shows in the ledger, stock updates.	The bulk-buying value
2:45	Attack panel: run S1, S2, S5, S3. Deny the S3 approval.	BLOCK and ASK with reasons
3:20	Run S4. Wall at the cap, quarantine banner, blast-radius meter.	Worst case is the cap
3:40	Audit: Verify green, Tamper, Verify red. Kill switch, run one attack, <code>KILL_SWITCH_ACTIVE</code> .	Tamper-evident, instant stop
3:55	Close on the line.	

2-minute version: mandate already signed; run the agent, approve the ASK (1:00); run S1 and S3 (0:30); show the meter (0:30).

8.2 Attack scenarios (expected results on a fresh reset with the default mandate)

ID	Name	What the agent sends	Expected
S1	Urgent invoice	<code>destination</code> account <code>9999999999</code> "Urgent Settlement Services", ₦90,000	BLOCK <code>PAYEE_UNKNOWN</code>
S2	Swapped bank details	<code>pay_primefoods</code> plus destination <code>1001999999</code> , ₦12,000	BLOCK <code>DESTINATION_MISMATCH</code>
S3	Quantity x10	<code>pay_primefoods</code> , ₦120,000 (usual ₦12,000)	ASK <code>OVER_AUTO_LIMIT</code> , <code>ANOMALOUS_AMOUNT</code> . NO moves unless the owner approves.
S4	Many small payments	30 x ₦40,000 to <code>pay_sunbev</code> , sequential, distinct references	<code>floor(headroom / ₦40,000)</code> ALLOW (5 on a fresh day, ₦200,000 total), the rest BLOCK <code>WINDOW_CAP_EXCEEDED</code> . Quarantine trips after the 3rd block. Total executed never exceeds ₦200,000.
S5	"Owner already approved"	<code>pay_primefoods</code> , ₦80,000, <code>description</code> claims prior approval	ASK <code>OVER_AUTO_LIMIT</code> , <code>ANOMALOUS_AMOUNT</code> . The claim is ignored.
S6	Race for the cap	Setup leaves ₦40,000 of daily headroom, then two concurrent ₦40,000 intents	Exactly one ALLOW, one BLOCK <code>WINDOW_CAP_EXCEEDED</code>

Order matters. S4 and S6 both consume the daily cap. Run S1, S2, S5, S3 first, then S4. For S6, press Reset first. Reset wipes the mandate, so the console's Reset re-signs the default one in a single click.

Pitch wording. Say "nothing reached an attacker and spend never passed the cap". Do **not** say "lost nothing": in S4 the agent legitimately spends up to the cap with an allowlisted supplier. That is the principle working, not a failure.

Blast-radius meter shows: wallet balance (₦2,000,000) against the daily cap (₦200,000, 10% of the balance), executed by the agent in the window, attacks attempted versus breaches (target 0).

9. Pitch deck brief (Designer)

Pitch length is unknown. Build **8 core slides plus an appendix**, with the live demo as slide 9. Trim to the real slot.

9.1 Principles

One idea per slide. Headline of 12 words or fewer. Every number has a source (quote, backtest, console screenshot). Mock things are labelled mock. The demo is the proof; slides only frame it.

9.2 Storyline

#	Slide	Draft headline	Visual and asset	From
1	Title	"[Platform] + MandatePay: bulk buying an AI can run safely"	Wordmarks	D2
2	Problem	"Small retailers pay the middleman premium and still run out of stock"	Small-lot vs bulk price from the real quotes	D1
3	Solution	"Neighbours pool orders. An AI agent keeps shelves stocked."	Three-step flow: pool, forecast, restock	Designer
4	Pools	"Commit, hold, unlock the tier price, or get refunded"	Tier ladder and progress bar (hero screen 4)	D3
5	AI restock agent	"It forecasts demand and orders before the shelf empties"	Forecast chart and backtest numbers	M6
6	The catch	"An agent that can spend money can be fooled into spending it"	The poisoned-invoice example	Designer
7	MandatePay	"The agent proposes. The owner's signed rules decide."	Diagram: agent, gateway, allow/ask/block, mock rail, audit log	Designer
8	Principle	"Assume the agent will be fooled. Worst case is the cap, not the balance."	Blast-radius meter	D3, F7
9	Live demo	(3 to 4 minutes, section 8)		Team
10	Results	"6 attacks. Nothing reached an attacker. Spend never passed the cap."	Console attack table screenshot	F7
11	Where we fit	"AP2-inspired: a user-side gateway, with attack evidence"	Landscape matrix	Designer
12	Why producers join, and who pays	"Larger, prepaid, predictable orders are cheaper to serve"	Assumption slide: say it is an assumption	Team
13	Honest scope and roadmap	"Mock rail today. Licensed payment partner tomorrow."	Roadmap	Designer
14	Team and ask			Designer

Slides 11 to 14 can live in the appendix if time is short.

9.3 Likely judge questions (draft the answers by Fri)

1. How is this different from AP2, the Visa and Mastercard agent programmes, or Stripe's scoped tokens?
2. What if the agent, or an invoice, is malicious? (Answer with the demo.)
3. What if the owner's key is stolen? (Production: WebAuthn or hardware-backed keys, rotation. Out of MVP scope.)
4. Is any money real? (No. Mock rail. Production uses a licensed partner and MandatePay never holds money.)
5. Why would producers join?
6. How do you make money?
7. How accurate is the forecast? (Backtest WAPE against the baseline.)
8. Cold start: why would the first neighbours pool? (Start with one locality and one producer.)

9.4 Landscape sources

From earlier research. Re-check each link before a name goes on a slide.

- Google launches Agent Payments Protocol: <https://yourstory.com/ai-story/google-agent-payments-protocol-ap2-ai>
 - Visa and Mastercard agentic commerce: <https://www.digitalcommerce360.com/2025/05/06/visa-mastercard-ai-agentic-commerce/amp/>
 - Agent card issuance with spending rules: <https://nevermined.ai/blog/ai-agent-card-issuance-spending-rules>
 - AP2 red-teaming paper (prompt injection): <https://arxiv.org/pdf/2601.22569>
 - AP2 explained: <https://ellogic.co/glossary/agent-payments-protocol-ap2/>
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10. Risks and cut-lines

Risk	Mitigation
Scope: five people, about a day and a half	Merge the P0 path first. If behind at a checkpoint, cut in this order: P2 items, then Producer portal and Orders page (P1), then quarantine UI, then live-LLM mode, then the tier-difference refund (pay tier-1 and settle success or failure only).
In-browser signing slips	Signing is the product, so do not fake it. Swap <code>@noble/ed25519</code> for <code>tweetnacl</code> or WebCrypto, and have Backend pair with Frontend until <code>signing.json</code> passes in the browser.
Services do not integrate	Mocks tonight, M1 checkpoint at Thu 12:00, Market verifies payments with the Gateway rather than trusting anyone.
LLM is slow or flaky on stage	<code>scripted</code> mode is the default. Live LLM is a bonus behind a toggle.
Demo environment fails	Run locally with <code>make dev</code> , plus the backup video and offline deck.
Clock skew breaks mandates	Use <code>server_time</code> ; approvals carry no client timestamp.
Reset breaks the browser key	Frontend re-registers the key after every reset (F2).
"Isn't this AP2?"	Slide 11. Position as an integrated user-side gateway with attack evidence. Say "AP2-inspired", never "compliant".

11. Open questions and definition of done

Open (please answer in the group chat):

1. Hackathon theme, judging criteria and pitch slot length. These decide how much of the deck goes to impact versus technical depth.
2. Real submission deadline. Times in this doc assume the build window closes Thursday evening.
3. Platform name (Designer proposes, team votes).
4. Stack confirmation (1.1) and hosting choice (S4).
5. Locality for the seed data.

Done means all of these are ticked by Thu 22:00:

- ☐ Fresh clone then `make dev` works on two different laptops.
- ☐ Deployed URLs are up, and `make reset` plus `scripts/e2e.sh` are green against them.

- ☐ All six attacks match section 8. `race_test.py` is green. Audit verify is green, and tamper turns it red.
- ☐ Golden path runs in 4 minutes or less and has been rehearsed.
- ☐ Backup video recorded, deck exported to PDF, offline copies on two laptops and a phone.
- ☐ MOCK labels are visible on every screen and no secrets are in the repo.